

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech Sem – II University Examination, 2010-11

PY – 101 Engineering Physics

Date: 14.05.11, Saturday

Time: 01:30 p.m to 04:30 p.m

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Indicate clearly, the option you attempt along with its respective question number.
4. Draw neat figures wherever required.

SECTION – I

- Q. 1** (a) Define reverberation and reverberation time. Why reverberation time always has to be maintained at its optimum value? 3
- (b) Explain the principle of propagation of light waves through an optical fiber. 4
- Q. 2** (a) A student measures three lengths a , b in cm , c in mm and a time t in $second$, where: $a = 50 \pm 4$, $b = 20 \pm 3$, $c = 60 \pm 5$, $t = 2.1 \pm 0.1$. Calculate: $a + b + c$, $(b - c) / t$, $(a + c) / t$ and square root of $(a \times c)$. 4
- (b) Define bel & decibel. What is the resultant sound level when a 70 dB sound is added to 80 dB sound? 5
- (c) Explain the determination of ultrasonic wavelength using acoustic grating method. 5

OR

- Q. 2** (a) Explain common sources of error in physics laboratory experiments with suitable examples. 4
- (b) Derive an expression for acceptance angle for an optical fiber. How it is related to numerical aperture? 5
- (c) Establish the relation between Einstein's coefficients A and B. 5
- Q.3** (a) What are ultrasonic waves? Explain with neat circuit, the generation of ultrasonic waves using piezoelectric oscillator. 7
- (b) Discuss briefly the conditions required for good acoustics. 7

OR

- Q.3 (a) What is population inversion? How it is achieved? Find the ratio of population of the two energy states in a laser where the room temperature transition between which is responsible for the emission of photons of wavelength $698.3 \times 10^{-9} \text{ m}$. Use $h = 6.625 \times 10^{-34} \text{ J s}$. 7
- (b) What is the principle of laser? Explain the construction and working of a solid laser, with energy level diagram. 7

SECTION - II

- Q.4 (a) Calculate the ratios $d_{100} : d_{110} : d_{111}$ simple cubic structure. 3
- (b) Define the term superconductivity? Write any three applications of superconductors. 4
- Q.5 (a) What is Hall effect? Explain briefly the experimental determination of carrier concentration and mobility of semiconducting specimen. The R_H of a specimen is $3.66 \times 10^{-4} \text{ m}^3/\text{C}$ and resistivity is $8.93 \times 10^{-3} \Omega \text{ m}$. Find μ and n . 7
- (b) How the physical and chemical properties of nanomaterials vary with their size? 7

OR

- Q.5 (a) Explain Meissner effect. Compare type-I and type-II superconductors on the basis of Meissner effect. 7
- (b) What are nanomaterials? Explain. Write any five applications of nanomaterials. 7
- Q.6 (a) Explain Bragg's law of X-ray diffraction. Monochromatic X-rays of $\lambda = 1.5 \text{ \AA}$ are incident on a crystal face having an interplanar spacing of $1.6 \text{ \AA}$. Find the highest order for which Bragg's reflection maximum can be seen. 7
- (b) Explain the terms: (i) crystal lattice, (ii) basis, (iii) unit cell, (iv) lattice parameter, (v) primitive cell, (vi) coordination number and (vii) atomic packing factor 7

OR

- Q.6 (a) What are Miller indices? What are the important features of Miller indices of crystal planes? Sketch the following planes of a cubic cell: (010), (110) and (111) 7
- (b) Discuss any seven features or properties of superconductors. 7